MDDB. Molecular Dynamics Data Bank. The European Repository for Biosimulation Data

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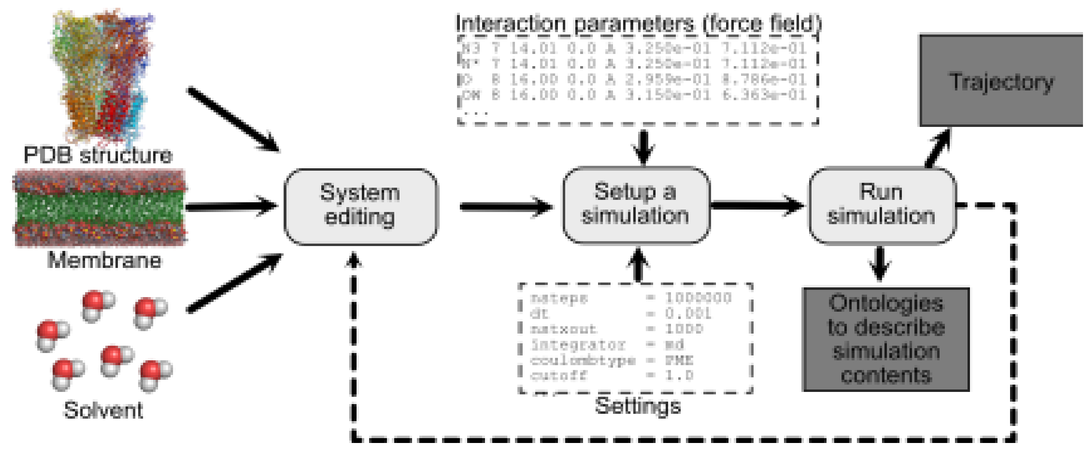
Sparse initiatives to build trajectory repositories have encountered several difficulties: i) the lack of trust of the community in the reliability of the data deposited; ii) the lack of interoperable standards and simulation ontologies; iii) uncertainties regarding the database technology required; v) difficulties of the users to interact in an open manner with the data; and vi) disconnection of the MD-field with neighboring communities.

The MDDB project intends to design a European-scale repository of MD simulation (and associated analysis), which will: i) optimize computational resources; ii) favor the analysis (and meta-analysis) of trajectories for many different perspectives and fields; iii) guarantee a fast and efficient interchange of information between groups; and iv) facilitate the integration of the MD simulation field into biological data infrastructures. The overall result will be a more efficient use of MD and the integration of the MD field into mainstream biology and chemistry research.

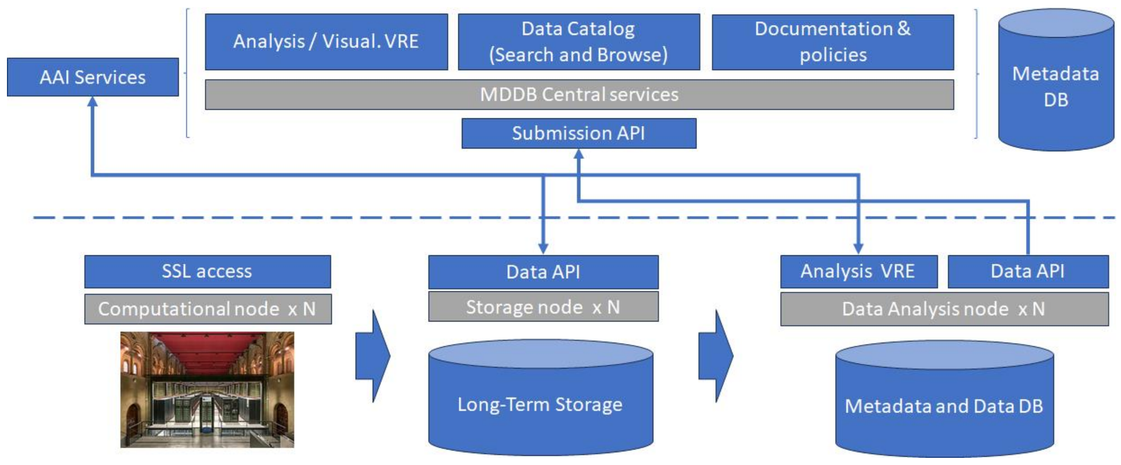
Data standardization and management



Our approach:

- Define a new common, **highly compressed data format** for trajectories storage and transmission.
- Address the **data quality analysis**, defining **data validation** and management procedures.
- Build data and **metadata ontologies** to facilitate the description of simulations.
- Define and **promote best practices for data generation** and storage.
- Define the appropriate data models to facilitate the **integration with the biological data ecosystem.**
- Address the **trust issue** by working with the community to enable embedding of **strict provenance information** in simulation trajectories.

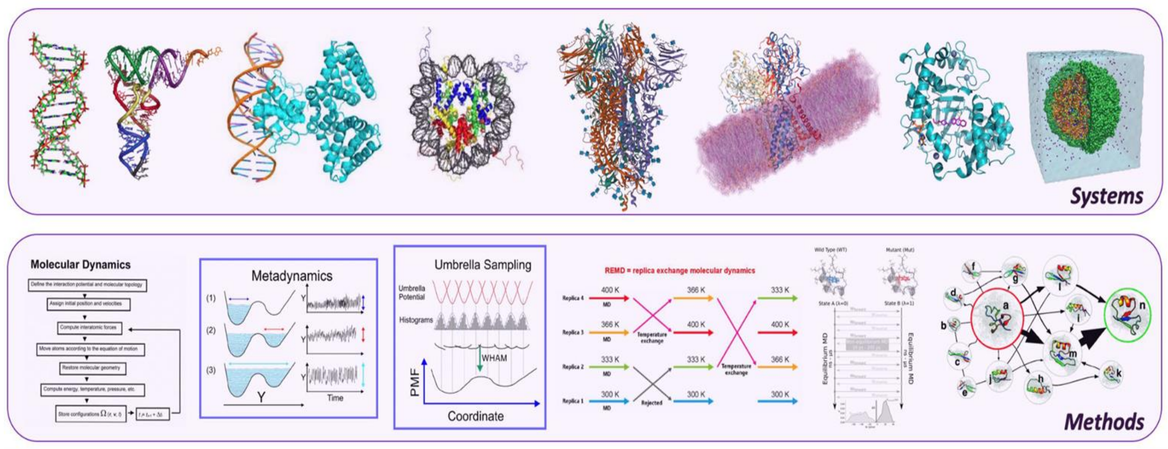
Technical infrastructure



MDDB is conceived as a **federated repository** including:

- **Central federation node**: manages simulation metadata, search and browse facilities, identifiers, data provenance, and main user interfaces.
- **Local data nodes**: hold datasets from specific scientific studies, maintained by data providers, limited computational capabilities.
- **Long term storage nodes**: provide archival storage with limited computational resources allowing data retrieval on demand but unable to perform analysis.
- **Computational & data nodes**: offers computational resources for analysis and simulation, along with limited long-term data storage.
- **HPC centers**: provides resources for simulation and analysis without long-term storage.

Pilot projects



[1] Hospital et al. 2016, [2] Newport et al. 2019, [3] Beltran et al 2024

Pilot projects have been selected to study the **technical feasibility** and **impact** of the **MDDB prototype**. The expertise gained will guide future database development, specifically to:

i. **test and refine** the different **methodologies** in real scenarios
ii. **attract the community** by demonstrating its benefits for academia and industry
iii. show the external communities the **power of MD trajectories** when organised in a **FAIR** manner

Existing data repositories created and maintained by MDDB partners were selected as **use cases**, including a variety of methods and systems:

BigNASim2.0[1]	**MemProtMD**[2]	**CNS-MD**	**ModTox**
COVID-19[3]	**Pyruvate Kinase**	**hexABC**	

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